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DataEcho: Reflective Telemetry Replay for Studying Navigation Strategy Refinement in XR

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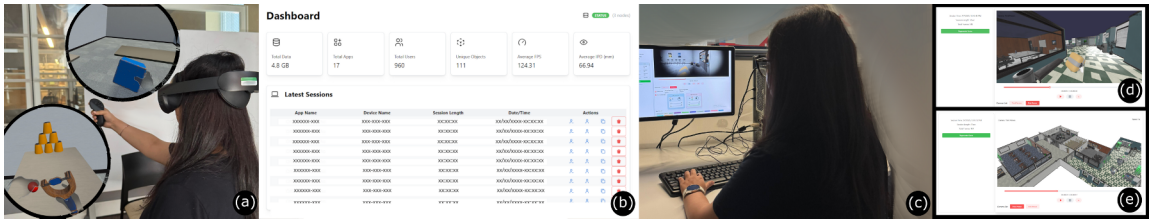


Figure 1: XR application using DataEcho data capture (a) streaming telemetry to back-end storage (b), allowing inspection via web replay for expert users (c) as well as integration into workflows for experimenters and subjects providing 1st-person (d) and 3rd-person (e) review.

Abstract

We investigate how persistent telemetry and reflective replay can support navigation strategy refinement in spatial learning tasks in XR. Our approach captures navigation traces, including trajectories, gaze direction, controller input, and interactions, through a Unity package and exposes them in a lightweight browser-based 3D replay interface, enabling post-hoc review without immersive hardware. We evaluated replay-review (interactive 3D replay with gaze visualization) against a no-review control in a controlled study (n=25) with two VR navigation sessions separated by one day. Replay-review led to a significant increase in continuous movement during the second session (greater decisiveness), while backtracking and head-orientation stability showed directional improvements; path optimality did not reliably improve in our low-redundancy layout. These results characterize replay-review as a bundled learner-facing reflection intervention and motivate follow-up studies that isolate the contributions of review baselines and individual replay features. This paper contributes: (1) DataEcho, a platform-agnostic telemetry pipeline supporting secure capture, scalable storage, and browser-based 3D replay of navigation sessions with gaze-ray visualization; (2) a controlled study evaluating replay-review (interactive 3D replay with gaze visualization) versus no review for VR navigation strategy refinement; and (3) limitations and directions for follow-up studies that isolate review baselines and individual replay features.

Index Terms: Extended Reality (XR), Behavioral Telemetry, Reflective Replay, Spatial Navigation, Immersive Analytics, Gaze Visualization, Learning Support Interfaces

1 Introduction

Immersive technologies are increasingly used for education, training, and simulation, where learners must navigate complex environments, allocate attention, and refine decision-making strategies. However, understanding how such strategies emerge and how they can be supported after task completion remains a central challenge for XR research. Common evaluation approaches, such as live observation or 2D video review, provide limited insight with an individual perspective, rarely allow learners themselves to revisit their own performance, and scale poorly for post-hoc analysis.

A key component of many XR training scenarios is *spatial learning*, defined as the acquisition of knowledge about environmental structure, including landmarks, routes, and decision points [8]. In navigation tasks, spatial learning is often reflected in more efficient movement, reduced hesitation at choice points, and improved anticipation of transitions [15]. We use the term *spatial strategies* to describe the behavioral patterns users adopt during navigation, such as decisiveness in route commitment, efficiency between wayfinding segments, and stability of attentional focus. These behaviors motivate our investigation of replay as a potential scaffold for navigation-based learning.

By separating immersive task execution from lightweight web-based review, learners and researchers can revisit sessions without specialized hardware and inspect where users moved and looked. This framing aligns with mixed-reality theory [21, 28], suggesting that learning and reflection can occur outside immersive displays when key behavioral cues are preserved. Several recent systems address XR telemetry capture and replay, demonstrating the maturity of research infrastructure but focusing largely on researcher-facing analysis rather than learner-facing reflection (e.g., [29, 13, 17]).

Gaze in particular has proven predictive of relevance in immersive sensemaking [34, 30], motivating its inclusion in reflective replay. At the same time, research has shown that motion traces themselves can be identifying [18, 1], underscoring the need for encrypted, privacy-preserving telemetry pipelines when persisting user data.

In this paper, we present both a framework and a study. Our system, DataEcho, captures trajectories and gaze during VR navigation tasks and provides a browser-based 3D replay interface. We then report a controlled study comparing navigation performance with and without reflective replay. Twenty-five participants completed two VR navigation sessions separated by a delay. Results show a significant improvement in continuous movement in the replay condition, with directional trends toward reduced backtracking and increased head-orientation stability; path optimality did not reliably improve in our low-redundancy layout.

This work makes the following contributions:

1. A platform-agnostic telemetry framework that supports secure capture, scalable storage, and web-based replay of navigation

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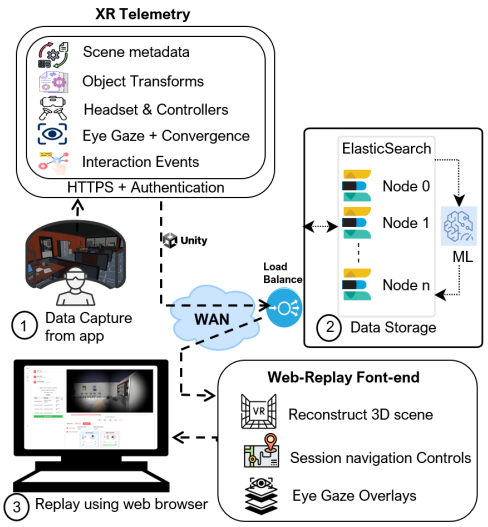


Figure 2: Overview of DataEcho's architecture including capture, storage, and replay layers.

data with gaze augmentation.

2. A controlled study evaluating reflective replay as a support for navigation strategy refinement in spatial learning tasks, showing that learner-facing replay review increases decisiveness during navigation, with directional trends toward reduced backtracking and greater head-orientation stability compared to no review.

3. Empirical characterization of replay-review as a bundled learner-facing reflection intervention, motivating follow-up studies that isolate review baselines and individual replay features.

Research Question and Hypothesis. We investigate whether access to asynchronous, learner-facing interactive 3D replay improves navigation behavior compared to no review. Specifically, we hypothesize that replay-review will lead to more decisive and less error-prone navigation on a subsequent session, reflected in reduced backtracking, increased decision confidence, and directional changes in scene-level attentional stability.

2 Related Work

Our work builds on four intertwined research areas: (1) visual representations and analytics in immersive environments, (2) XR telemetry systems for capture and replay, (3) gaze, reflection, and cognitive scaffolding, and (4) mixed-reality review and ethical considerations.

2.1 Visual Representations for Cognition and Training

Visualization in XR has long been recognized as a means of amplifying human cognition and supporting situated learning. Prior work has shown that immersive representations can promote conceptual understanding and skill acquisition across domains such as education [16, 12] and training [5, 32, 33].

In contrast, substantially less attention has been given to *post-hoc reflection*—how learners revisit their own actions and attention to refine strategies over time. Research on structured debriefing shows that reviewing one's own actions yields measurable performance gains compared to passive video review [27, 23, 22]. We extend these principles to XR by treating persistent telemetry and interactive 3D replay as learner-facing scaffolds for reflective practice beyond the moment of interaction.

2.2 XR Telemetry, Capture, and Replay Systems

In parallel, XR research has produced a range of telemetry frameworks supporting experiment orchestration, capture, and replay. Systems such as Ubiq-Exp [29] and UXF [3] emphasize reproducibility and experiment management, while PLUME [13], ISA [17], and PEARL [19] provide increasingly comprehensive capture and review pipelines.

As summarized in Table 1, these systems offer robust researcher-facing capabilities, including exhaustive logging, offline replay, and collaborative analysis. However, they are rarely evaluated as *learner-facing* reflection tools, and few empirically test how replay affordances (e.g., interactivity, perspective control, gaze overlays) relate to spatial learning in controlled studies. We address this gap by evaluating reflective replay as a cognitive scaffold for navigation behavior.

2.3 Gaze, Reflection, and Cognitive Scaffolding

Eye-gaze is a strong proxy for attention and strategy in immersive environments. Prior work shows that gaze predicts information relevance during immersive sensemaking, highlighting its potential for guiding reflection [34, 31]. By making attentional dynamics explicit, gaze overlays support reflective diagnosis of hesitation and misallocation of focus.

Our approach incorporates gaze into interactive replay not as a standalone signal, but as part of a broader reflective scaffold. While the present study evaluates replay that includes gaze visualization, isolating the unique contribution of gaze relative to interactive replay remains an important direction for future work.

2.4 Mixed Reality Review and Ethical Considerations

The separation of immersive task execution from non-immersive reflective review aligns with mixed-reality theory. Milgram and Kishino's reality–virtuality continuum [21], later expanded by Skarbez et al. [28], suggests that meaningful learning can occur outside immersive displays when critical perceptual cues are preserved. Our web-based replay approach operationalizes this principle by enabling reflection on spatial and attentional traces in a lightweight browser-based environment.

At the same time, persistent XR telemetry raises ethical concerns, as motion traces can function as behavioral biometrics [26]. These risks underscore the responsibility of systems that retain user data. Our framework incorporates encrypted transport, pseudonymous identifiers, and access controls to balance the learning benefits of reflective replay with responsible data stewardship.

3 DataEcho Architecture

3.1 Overview

Existing XR telemetry systems support orchestration, capture, and replay for reproducible studies and collaborative analysis [29, 13, 17, 19, 3]. DataEcho is designed to repurpose telemetry as a learner-facing resource by enabling post-hoc, interactive replay on the Web, without requiring immersive hardware. This separation supports reflective review while allowing controlled studies that examine how replay affordances like interactivity, perspective control, and gaze overlays relate to spatial navigation behavior.

DataEcho consists of three primary components (cf. Fig. 2): a *data-capture layer*, implemented as a lightweight Unity package, a scalable *storage layer*, utilizing Elasticsearch [9], and a web-based *replay layer*, providing interactive visualizations via React [20] and Three.js [6].

3.2 Data Collection

DataEcho integrates into Unity as a lightweight package that captures XR telemetry via a configurable component (cf. Fig. 2). Capture parameters (e.g., sampling interval, tracked objects, and storage credentials) are specified through the Unity Inspector without

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System	Platform Support	Capture Scope	Storage	Replay Features	Multi-User Capture	Data Sharing	Gaze Integration	Open Source	Learner-Facing Replay
VREX / RePlayAR	Unity	Scene states, partial inputs	Unity local logs	Scene reenactment in VR	✗	✗	✗	✗	✗
Ubiq-exp	Core Ubiq	Pose, avatars, events	✓ *(file)*	3D trajectory, live monitor	✓	✗	✗	✓	✗
Plume	Windows only	6DoF trajectories	✓ *(file)*	3D trajectory	✗	✓ *(file)*	✓	✓	✗
Kinematic Maze	Web *(not released)*	✗ *(datasets)*	✗	prototype *(not released)*	✗	✓ *(file)*	✗	✗	✗
DataEcho *(Ours)*	Web *(Cross-platform)*	Pose, gaze, controllers, events	Elasticsearch (scalable DB)	3D trajectory, gaze & input	✓	✓ (API)	✓	*~upon publication*	✓ *(Shareable URL)*

Table 1: Comparison of XR telemetry and replay systems across major differentiators.

modifying application logic, enabling reuse across applications and studies. In each collection cycle, the logger records object pose, controller input, interaction events, and eye-gaze data, along with minimal scene and session metadata. When available, gaze direction is logged at the headset update rate (upto ∼90Hz) with invalid samples (blinks/tracking loss) omitted; replay renders a world-frame gaze ray via raycast per sample without fixation estimation.

Telemetry frames are serialized and asynchronously uploaded to an Elasticsearch backend using the ND-JSON Bulk API [10]. Data are sampled at a user-defined interval (typically ∼10 ms) and transmitted in small batches (up to 20 frames) to reduce overhead and latency.

3.3 Storage and Security

DataEcho stores telemetry data in Elasticsearch, enabling scalable indexing and retrieval of time-stamped spatial and interaction data. DataEcho incorporates privacy-preserving design principles, including encrypted transport, role-based access control, and pseudonymous session identifiers. No personally identifiable information is stored, and replay access can be restricted via time-limited, tokenized URLs.

3.4 Web-Replay Front-end

DataEcho provides a lightweight, browser-based 3D replay interface that reconstructs immersive XR sessions from logged telemetry without requiring immersive hardware. The replay preserves the spatial and temporal structure of each session, enabling participants and researchers to revisit prior navigation and interaction behavior post-hoc.

The interface (cf. Fig. 3) supports interactive session inspection through time manipulation, viewpoint switching, and path visualization, allowing users to explore trajectories and object interactions within the reconstructed scene. For sessions with eye tracking, gaze overlays and controller-input visualizations are displayed to contextualize attention and action during navigation. In the current implementation, replay sessions can be recorded as video through the interface, while structured raw telemetry (JSON) remains accessible through the backend for analysis and archival.

To support asynchronous review, sessions can be shared via time-limited guest URLs protected by token-based access control [14]. Guest access (cf. Fig. 1d & Fig. 1e) provides essential replay functionality used in this study (timeline navigation, perspective control, and gaze overlays), while advanced tools (e.g., video recording, model assignment, session events table) remain restricted to authenticated researchers [1] (cf. Fig. 3).

4 Experiment

4.1 Objective

Our objective is to evaluate *reflective replay* as a cognitive scaffold for navigation strategy refinement in spatial learning tasks. We test whether asynchronous, learner-facing 3D replay with gaze leads to more decisive navigation on a subsequent session compared to no review, using DataEcho solely as the enabling capture-and-replay mechanism.

In this task, we operationalize spatial learning through measurable improvements in how accurately and efficiently participants traverse the environment's structure. Path performance is captured by the *Path Efficiency Index (PEI)* (walked vs. optimal distance). Strategy refinement is captured by the *Backtracking Index (BI)* (route reversals/detours) and *Confidence Index (CI)* (continuous movement during navigation). Head-orientation stability (FF) serves as a proxy for attentional stability, defined as the fraction of consecutive head-direction samples with angular change below 0.5°. Together, these measures are intended to capture both outcome-level performance and intermediate behavioral dynamics that may change after replay, even when shortest-path efficiency may be less sensitive in low-redundancy indoor layouts.

4.2 Task Design

Our study was carried out within a virtual office environment [11] created in Unity, usable on Android headsets, with gaze overlays available on eye-tracked devices (cf. Fig. 4). We tested on a Meta Quest Pro, chosen for its integrated eye-tracking capability, which was enabled for all participants. We used two-thumbstick locomotion, mapping left-stick input to translation (fixed maximum speed) and right-stick input to yaw rotation (fixed turn rate).

Participants completed two single-trial navigation stages separated by one day. In each stage, participants started at the same spawn location, picked up a single virtual package from a fixed

[1] Source code for the DataEcho Unity package and web replay system will be released upon publication.

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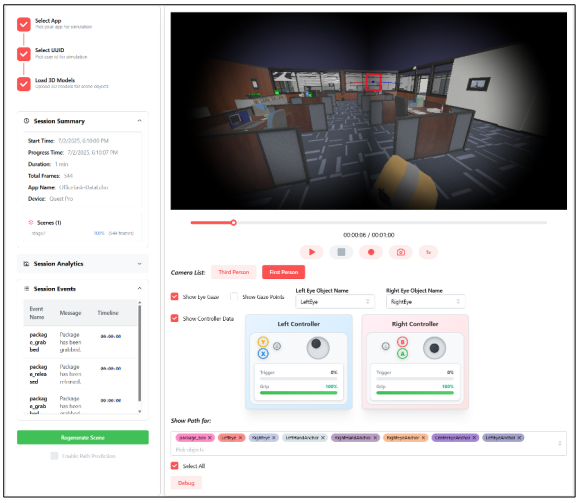


Figure 3: Screenshot of DataEcho's expert replay interface showing additional options like video recording, session events, and VR controllers input.

Measure	Replay	Control
Age (years), M±SD	27.50±3.90	25.69±3.60
Gender (F/M)	3/9	2/11
VR experience (1–10), M±SD	3.92±2.54	2.85±2.34

Table 2: Participant demographics and self-reported prior experience by condition (Replay $n = 12$, Control $n = 13$). Experience ratings used 10-point scales (1 = none, 10 = expert).

desk, and delivered it to a designated drop-off location (cf. Fig. 4), marked in-world by a green placeholder marker (cf. Fig. 5). A stage ended when the package was placed within the drop-off trigger. To support the analysis of progress along the route, we placed 16 invisible checkpoint colliders at predefined wayfinding transitions (i.e., each doorway and corridor junction) (cf. Fig. 4) identical across participants and registered when the HMD entered the collider volume.

4.3 Participants and Study Design

Twenty-eight participants were recruited. After completing Stage 1, participants were randomly assigned to either the replay group or the control group. 3 participants were excluded due to data quality issues, resulting in a final analyzed sample of $n = 25$ (Replay $n = 12$, Control $n = 13$). Participant demographics and self-reported prior VR experience are summarized in Table 2. The experiment setup constitutes a 2×2 mixed factorial design, with *Group* as the between-subjects factor and *Session* as the within-subjects factor. Eligibility criteria included age $\geq$18, normal or corrected-to-normal vision, and no history of severe motion sickness. No prior VR experience was required. Randomization was stratified to balance age, gender, and VR experience across groups. All procedures were approved by the IRB of University of Arkansas at Little Rock (IRB #25-038), and informed consent was obtained.

4.4 Procedure

4.4.1 Stage 1 (Day 1)

To ensure familiarity with smooth locomotion and object interaction, participants completed two brief warm-up tasks: (1) pick up virtual objects from one table and place it on an adjacent table, and (2) aim and launch a projectile at a target by pulling and releasing a sling-shot.

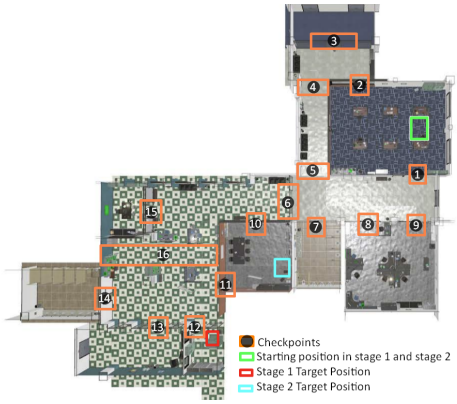


Figure 4: Virtual office environment used for the navigation task. Checkpoints were placed at key wayfinding transitions.

Participants then began the delivery task, which required grabbing a virtual package from a desk and delivering it to a specified destination (cf. Fig. 5). Each participant completed one navigation trial per stage. Throughout the task, telemetry data (head pose, controller input, and gaze) were streamed and stored via DataEcho. After completing the task, participants filled out a post-experience questionnaire consisting of the Simulator Sickness Questionnaire (SSQ [2]; standard SSQ item severity ratings) and a short set of additional Yes/No/Maybe items about comfort and task experience.

4.4.2 Review Phase (Day 1)

After completing Stage 1, participants were randomly assigned to one of two experimental groups: *Replay-review* participants were given time-limited (24-hour URL) access to an asynchronous, interactive 3D replay of their Stage 1 session via DataEcho (including gaze visualization), whereas *Control* participants received no review access. To verify interactive (vs. passive) replay use, the system logged camera/perspective manipulations and temporal navigation. These logs were used to derive replay engagement measures (active time, interaction counts, and replay events) reported in Sec. 5.5.

4.4.3 Stage 2 (Day 2)

On the following day, participants returned to complete a second delivery task from the same starting location, delivering the package to a new destination along a previously traversed route, requiring participants to reuse and refine prior navigation decisions.

4.5 Data Processing and Metrics

Of the 28 recruited participants, 3 were excluded due to tracking data loss exceeding 10% or premature termination, resulting in a final analyzed sample of n = 25. All metrics were computed separately for Stage 1 and Stage 2, with change scores Δ (Stage 2 – Stage 1) used for between-group comparisons.

Path Efficiency Index (PEI). For each trial, the shortest walkable route was obtained using Unity's NavMesh path solver [24]. Actual path length was computed by summing frame-to-frame HMD displacements during active locomotion (joystick magnitude > 0.1). PEI was defined as the ratio of actual path length to the NavMesh–optimal path length, $\text{PEI} = \frac{\text{actual}}{\text{optimal}}$, with PEI ≥ 1 indicating deviation from the shortest route.

Backtracking Index (BI). Instantaneous heading was estimated from the HMD forward vector projected onto the horizontal plane. A backtracking event was counted whenever the change in heading exceeded 150° relative to the preceding ∼1 s window.

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(a) 1st-person view at subject's entry location in all stages. **(b)** Package in subject's hand. **(c)** Stage 1 drop-off location marked by green placeholder object. **(d)** Stage 2 drop-off location marked by green placeholder object.

Figure 5: Office-navigation task: (a) entry, (b) grabbing the package, and delivery to the drop-off points in (c) Stage 1 and (d) Stage 2.

Similar reversal-style thresholds have been used in prior VR locomotion/navigation analyses based on heading dynamics [25]. BI was defined as the proportion of frames classified as backtracking, with ΔBI capturing changes across stages.

Confidence Index (CI). CI is the proportion of locomotion frames in a trial (higher CI indicates more continuous movement). We use CI as a proxy for decisiveness because decision-making at intersections often involves reorientation and information seeking, and navigators may pause forward movement to look around before committing to a route [4]. Let u_t be the magnitude of the joystick locomotion input (`Primary2DAxis`) and apply a deadzone threshold $\tau = 0.1$ to ignore unintentional input noise; a frame is labeled as locomotion when $u_t > \tau$.

Head-orientation stability (FF). Rotational stability was derived from frame-to-frame changes in head-orientation. For each pair of successive samples, we computed the angular difference between normalized head-direction vectors. Following thresholding strategies inspired by angular stability fixation and saccade classification in VR [7], we applied angular-change bands (<0.5°, 0.5°–10°, >10° per sample) to categorize transitions as fixation-like, smooth-pursuit-like, or saccade-like. Head-orientation stability (FF) was computed as the proportion of small-change transitions per stage, with ΔFF representing the across-stage change.

5 Results

We report outcomes for the four telemetry-derived measures introduced in Sec. 4.1 and computed as described in Sec. 4.5: the Path Efficiency Index (PEI), Backtracking Index (BI), Confidence Index (CI), and Head-orientation stability (FF). We use Wilcoxon for within-group Stage1 vs Stage2, and Mann–Whitney U for between-group change scores. Across measures, the clearest between-group evidence is observed for CI, while PEI, BI, and FF provide contextual and directional support. All analyses use nonparametric tests and report effect sizes where applicable; bars show group means with error bars of ±1 standard error of the mean (SEM).

5.1 Path Efficiency Index (PEI)

In the replay group, mean PEI decreased from 2.38 in Stage 1 to 1.99 in Stage 2 (Wilcoxon $W = 22.0$, $p = .204$). The control group showed a similar reduction from 2.75 to 2.11 (Wilcoxon $W = 21.0$, $p = .094$). Although the control group exhibited a slightly larger mean improvement in PEI (ΔPEI_{ctrl} = 0.65 vs. ΔPEI_{replay} = 0.39), the between-group comparison of improvements was not statistically reliable (Mann–Whitney $U = 73.0$, $p = .807$, rank-biserial $r \approx 0.06$; cf. Fig. 6a). Given the low-redundancy layout, PEI is treated primarily as a coarse check for gross path efficiency, with BI and CI better capturing decision dynamics at choice points.

5.2 Backtracking Index (BI)

The replay group showed a reduction in BI from Stage 1 (M = 2.37%) to Stage 2 (M = 1.90%), whereas the control group showed a slight increase in backtracking (Stage 1: M = 1.86%; Stage 2: M = 1.94%). Within-group Wilcoxon tests did not reach statistical significance (Replay: $W = 18.0$, $p = .11$; Control: $W = 33.0$, $p = .67$).

To visualize improvement directly, Fig. 6b presents the distribution of change scores $\Delta BI = BI_{Stage2} - BI_{Stage1}$. Negative values indicate reduced backtracking (improvement). The replay group exhibited a median reduction of -0.46 percentage points, whereas the control group showed a slight increase ($+0.11$ percentage points). A between-group comparison of these change scores did not reach significance (Mann–Whitney $U = 52.0$, $p = .17$, rank-biserial $r = -0.28$), but shows a directional pattern consistent with our hypothesis: participants who engaged in replay tended to backtrack less on the second exposure, suggesting (but not proving) more stable spatial strategies. In contrast, the control group exhibited a slight increase in reversals, consistent with natural variability in unassisted spatial recall.

5.3 Confidence Index (CI)

The replay group showed an increase in continuous movement from Stage 1 (69.4%) to Stage 2 (74.3%) (Wilcoxon $W = 19.0$, $p = .129$), while the control group decreased from 75.3% to 69.6% (Wilcoxon $W = 29.0$, $p = .27$) (cf. Fig. 6c). The between-group difference in improvement was statistically significant (Mann–Whitney $U = 115.0$, $p = .047$, $r = 0.40$), indicating a moderate effect and representing the clearest behavioral impact of replay in this study. This suggests that users felt *less hesitant* and *more confident* after reviewing their prior performance, though the present design does not isolate which replay features contributed to this effect.

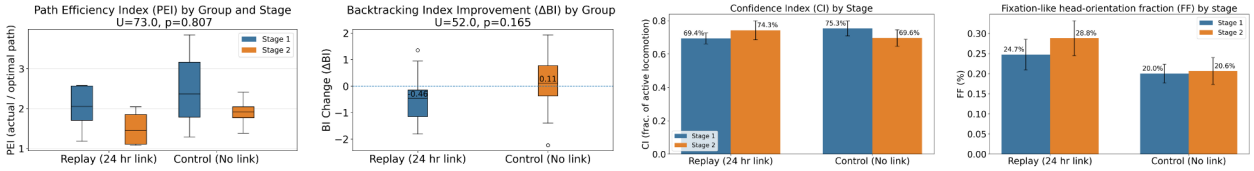
5.4 Head-orientation stability (FF)

In the replay group, FF increased from a mean of 24.7% in Stage 1 to 28.8% in Stage 2 (Wilcoxon $W = 15.0$, $p = .064$). In the control group, FF remained essentially unchanged (20.0% to 20.6%; Wilcoxon $W = 39.0$, $p = .685$) (cf. Fig. 6d). However, a between-group comparison of improvements did not yield a reliable effect (Mann–Whitney $U = 91.0$, $p = .497$, rank-biserial $r = 0.14$). Thus, FF exhibits a directional trend toward greater head-orientation stability in the replay condition, but the effect is not statistically conclusive. These patterns are consistent with reflective replay being associated with higher decisiveness (CI) and a directional increase in attentional stability (FF), but do not provide conclusive evidence for effects on FF.

5.5 Replay Engagement (Interaction Logs)

Replay engagement was quantified from interaction logs collected during the review phase (replay-review group, n=12) and aggregated across the 24-hour window. First review occurred at a median of 20.1 h after Stage 1 (IQR 2.28–22.92 h; range 0.62–23.03 h), and timing was not modeled as a predictor of performance change in this study. Navigation trials lasted approximately 68.2 s on average, while median active review time was 364.1 s (mean 540.9 s; range 100.0–1631.7 s). Adoption differed by control: 83.3% used camera controls, and 41.7% used timeline scrubbing. Replay events suggested repeated inspection (median 2.5 replay starts; median 6 replay stops). Exploratory Spearman tests related engagement to ΔBI

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(a) Path Efficiency Index **(b)** Backtracking Index. **(c)** Confidence Index. **(d)** Head-orientation stability.

Figure 6: Task performance and behavioral metrics: (a) Path Efficiency Index (PEI; Stage 1 vs. Stage 2). (b) Backtracking Index improvement ($\Delta\text{BI} = \text{BI}_2 - \text{BI}_1$; negative indicates reduced backtracking). (c) Confidence Index (CI). (d) Head-orientation stability (FF). Bars show group means; error bars show ± 1 SEM.

to help prioritize replay affordances for follow-up studies. Only camera control was associated with ΔBI ($r_s = 0.67$, $p = .016$); all other measures were not reliable ($p > .49$). Given the small sample size and multiple comparisons, these results are descriptive and motivate feature-isolation studies rather than causal claims.

6 Discussion

6.1 Reflective Replay as Cognitive Scaffold

Our results provide evidence that learner-facing reflective replay can support aspects of spatial learning in immersive navigation tasks, particularly those related to strategy refinement rather than global path optimization. Participants who reviewed their prior performance via interactive 3D replay showed increased decisiveness and a directional reduction in backtracking on a subsequent navigation attempt, whereas participants without review showed no comparable improvement. In contrast, improvements in shortest-path optimality, as measured by PEI, were limited for both groups.

Replay did not reliably shorten navigation paths, likely due to the low redundancy of the office layout and the constrained number of viable routes. However, replay-review was associated with fewer aborted routes and fewer heading reversals (BI) at decision points, suggesting that participants may have refined how they committed to routes rather than discovering globally shorter paths. In this study, BI captures moment-to-moment route commitment and decision reversals at junctions, which might change even with limited exposure, reflecting strategy refinement rather than long-term map acquisition. This distinction matters in indoor environments, where refinement often appears as reduced hesitation rather than shorter paths.

Replay interaction logs confirmed active engagement during review (Sec. 5.5), indicating replay functioned as exploratory reflection rather than passive viewing. We therefore interpret reflective replay as a cognitive scaffold that externalizes prior experience and supports post-hoc reasoning about decision moments.

6.2 Decisiveness and Attention

The replay group demonstrated a significant increase in decision confidence (CI), while the control group exhibited a decrease, indicating reflective replay supported more continuous movement with fewer micro-pauses, consistent with reduced hesitation during navigation. The expectation that reflective replay would improve both decisiveness and attentional stability was supported in part.

Scene-level attentional stability, estimated using head-orientation stability (FF), showed a directional increase in the replay group but did not reach statistical significance in between-group comparisons. While these trends are consistent with increased attentional steadiness during navigation, the present study was not powered to conclusively detect such effects. FF was intentionally derived from head-orientation dynamics as a scene-level proxy for attention allocation rather than fine-grained eye-in-head fixation classification. Overall, replay-review primarily affected decisiveness, with weaker evidence for attentional stability, consistent with strategy refinement.

6.3 Positioning Within XR Infrastructure

Our findings highlight an important distinction between telemetry collected *for research* and telemetry presented *for learning*. Existing frameworks [29, 13, 17, 19, 3] are primarily designed for researcher-facing analysis, reproducibility, and orchestration. In contrast, our results demonstrate that when telemetry is returned to learners in an interactive, inspectable form, it can function as a scaffold for strategy refinement with measurable behavioral effects.

Reflective replay should therefore be viewed not as a replacement for existing XR infrastructure, but as a complementary layer that transforms recorded telemetry into a learner-facing resource. This perspective reframes replay from a post-hoc analysis tool into an integral component of learning-oriented XR experiences.

6.4 Limitations

Several factors limit interpretation of these findings. The office environment afforded few truly divergent routes, reducing the sensitivity of PEI. We compared replay-review (interactive 3D replay with gaze) to no review, without another review baseline, so it remains unclear whether the benefit comes from having any review or from specific replay features. Replay engagement analyses were exploratory and underpowered ($n = 12$) with multiple comparisons, so correlations are descriptive rather than evidence of causal mechanisms. We did not collect standardized usability ratings of the replay interface; thus, user experience is characterized only via interaction logs (Sec. 5.5), and usability remains future work.

7 Future Work

Future work should isolate individual replay affordances to clarify their contributions to navigation learning. Controlled comparisons that vary perspective control, temporal navigation, and gaze visualization would identify which features drive gains in decisiveness and reductions in backtracking. More complex environments and longer-term studies are also needed to test durability and transfer of replay-supported learning. An important direction for future work is a direct comparison of 2D and 3D replay to examine their relative effects on spatial updating and navigation strategy.

8 Conclusions

This work examines how reflective telemetry replay supports navigation strategy refinement in immersive environments. We present DataEcho, a lightweight telemetry pipeline that enables browser-based 3D replay with gaze overlays. In a controlled study, replay review increased decisiveness in subsequent navigation, while backtracking and attentional stability improved directionally and route optimality (PEI) did not reliably improve in our low-redundancy layout.

These findings position 3D replay review as an *active instrument of reflection and metacognition* and suggest that learner-facing replay review (a bundled intervention combining interactivity, perspective control, and gaze augmentation) can yield measurable benefits for navigation behavior.

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